

MicroFit - Your very own Step Counter

Microbit

Beginner

10 Mins

1 Overview

Ever wondered how a smartwatch or fitness band counts your steps? In this project, you're going to build your very own step counter using a micro:bit!

Inside your micro:bit is a clever little sensor called an accelerometer. It can feel when the micro:bit moves. We'll use it to spot every step you take, add them up, and show the total on the LED matrix (the grid of lights on the front).

The best bit? When you're done, you'll have a real working fitness tracker you can pop on and take outside. Don't worry if something doesn't work first time, that's all part of coding. Let's give it a go!



What you'll learn

- Make and use your own variable
- Use the micro:bit's shake sensor to spot movement
- Show a number on the grid of lights
- Use the buttons to control your project
- Use a while loop
- Make decisions with if statements

2 What you'll need



What you'll need

- The micro:bit EduBlocks editor
- micro:bit
- micro USB cable
- battery pack for the micro:bit (optional)

3 Key words

Key words

Variable

A variable is something that can change. Think of it like a labelled box: you pop something inside, and the label tells you what it is. You can swap what's inside whenever you like.

A variable called score might start at 0, then change to 1, 2 and 3 as you collect points in a game.

While loop

A while loop keeps repeating its instructions for as long as something stays true. The moment that thing stops being true, the loop stops.

"While it's still raining, keep the umbrella up." As soon as the rain stops, you put the umbrella down, and the loop ends.

If statement

An if statement checks whether something is true, and only runs its instructions when it is. It's how code makes decisions.

"If it's cold, put on a coat." If it's warm, you skip the coat, the action only happens when the check is true.

Accelerometer

An accelerometer is a tiny part inside the micro:bit that senses movement, like tilting, shaking or being turned over. It turns that movement into numbers your code can react to.

A bit like the sensor in a games controller that knows when you tilt it to steer, the accelerometer notices when the micro:bit moves.

LED Matrix

The LED matrix is the grid of little red lights on the front of the micro:bit. By switching them on and off in patterns, your code can show numbers, letters and tiny pictures.

Like a mini scoreboard made of dots, the LED matrix lights up to spell out your step count.

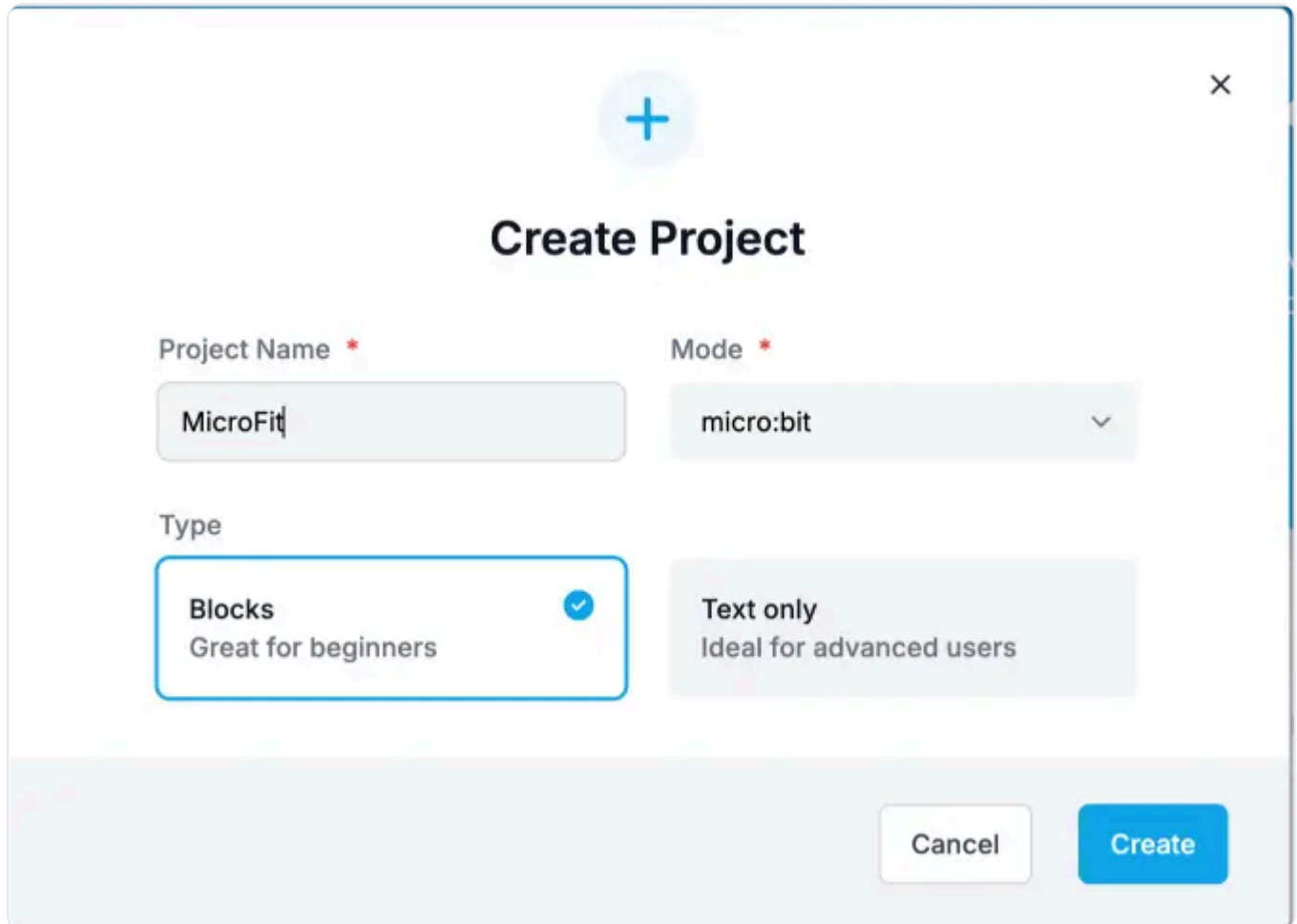
Button

A button is a switch you press to tell your project to do something. The code can check whether it's being pressed or not.

Pressing a doorbell is just like pressing a button, one press sends a signal that makes something happen.

4 Set up your micro:bit EduBlocks Editor

1. Open your web browser. We recommend Google Chrome or Microsoft Edge.
2. In the address bar type app.edublocks.org.
3. Under **Create New Project**, select **micro:bit**. Name your project **MicroFit**, and make sure **block** is selected under type.



Create Project

Project Name *

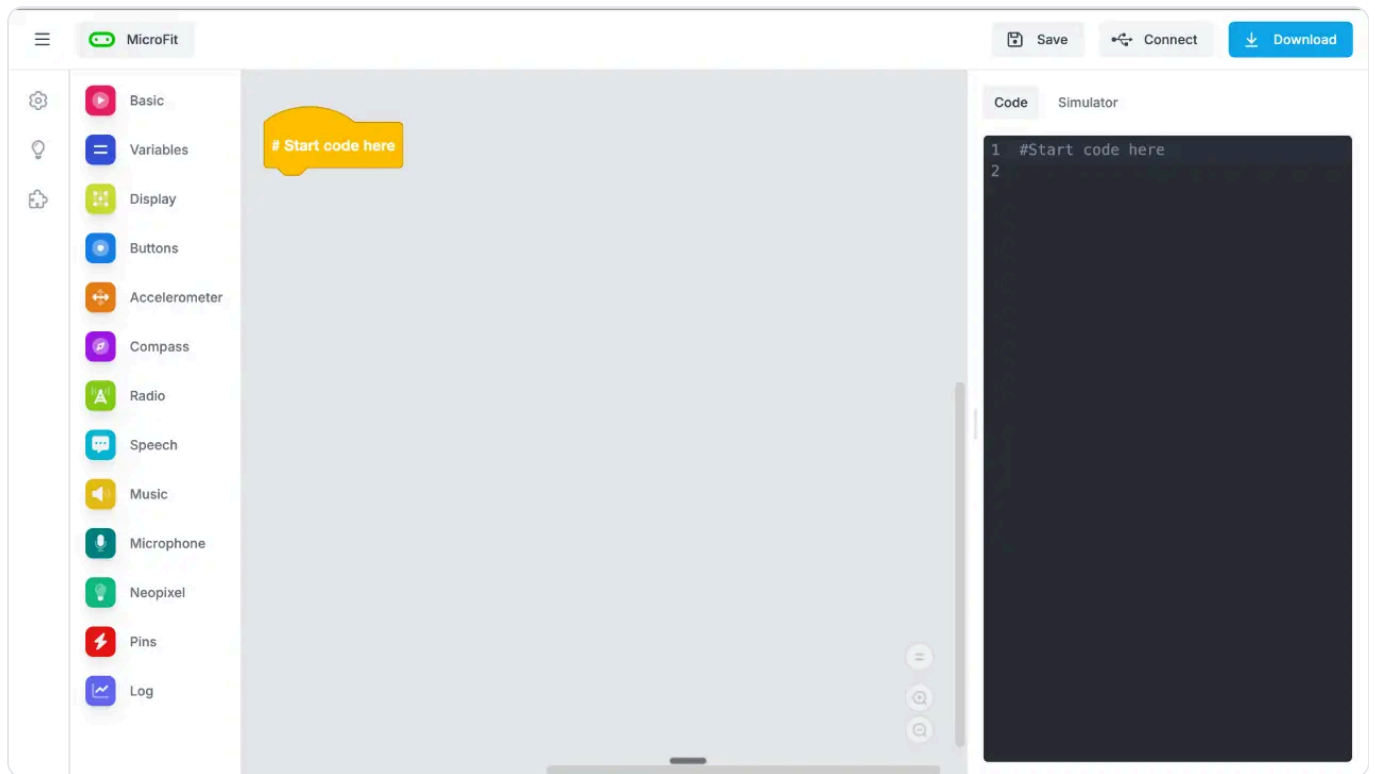
Mode *

Type

Blocks ☒ Great for beginners

Text only ☐ Ideal for advanced users

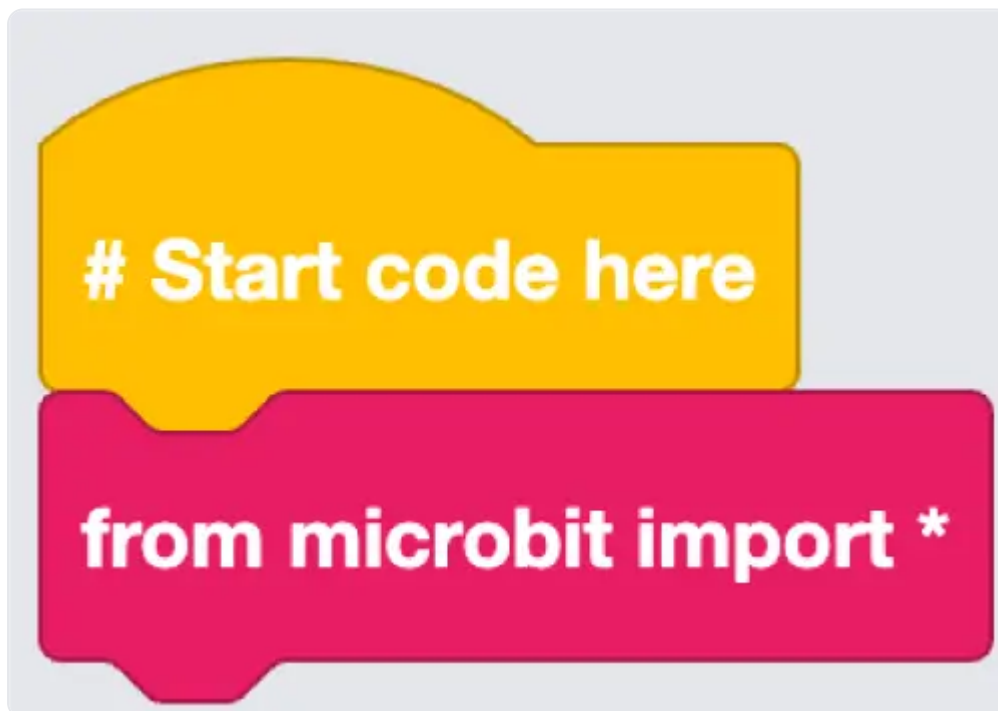
4. Select **Create** to open the micro:bit editor.



5 Import the MicroPython library

Select **Basic**, then **Imports**, then snap a `from microbit import *` block onto the `#start code here` block.

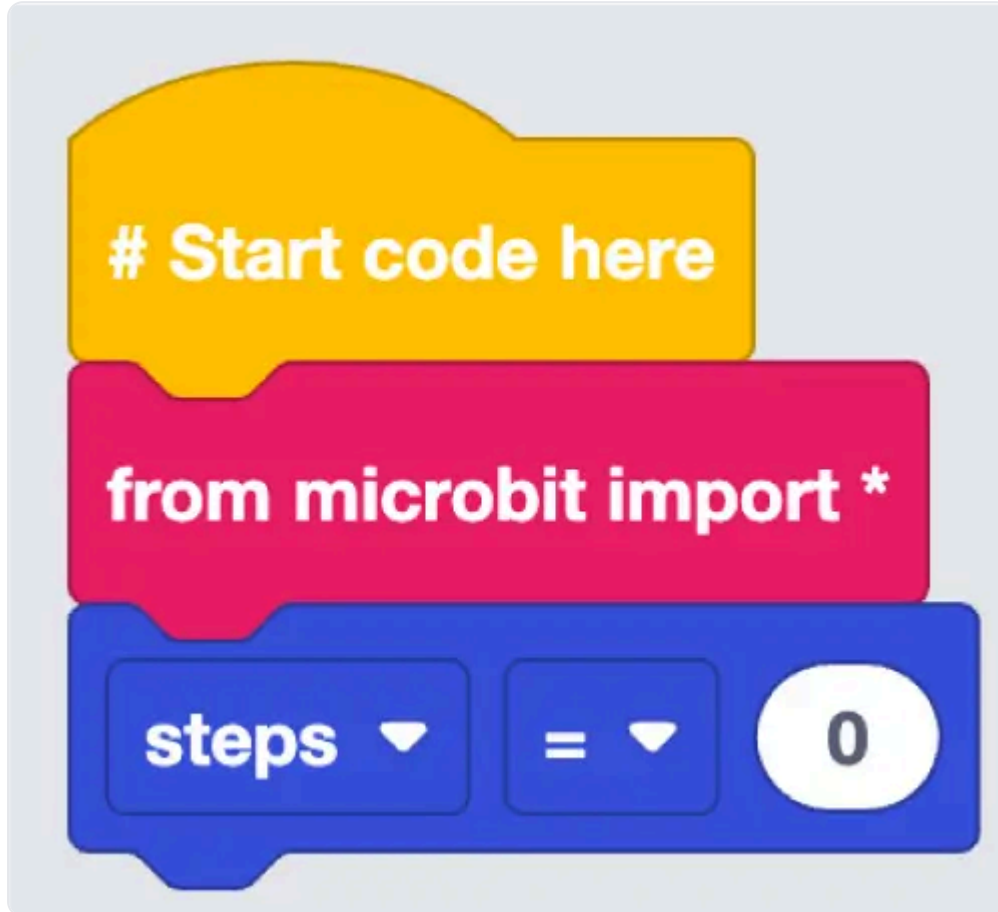
This imports the micro:bit library into EduBlocks so we can use all the micro:bit's functions.



6 Set steps to 0

1. Select **Variables**, choose **Create a Variable**, and name it **steps**.
2. Snap a `steps = 0` block to the `from microbit import *` block.

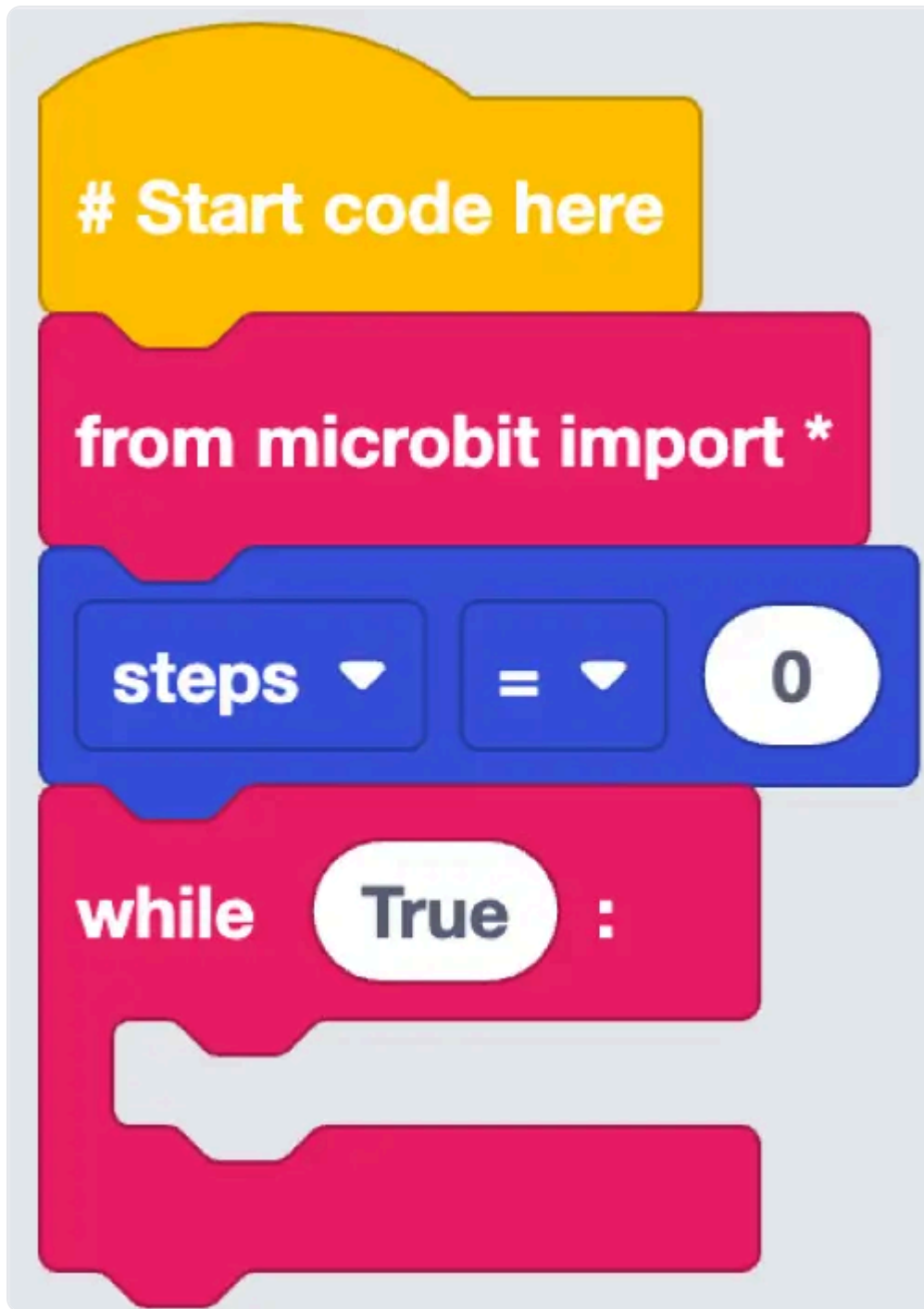
This makes a variable called **steps** and starts it at zero.



7 Create a while True loop

Select **Basic**, then **Loops**, then snap a `while True:` block onto the `steps = 0` block.

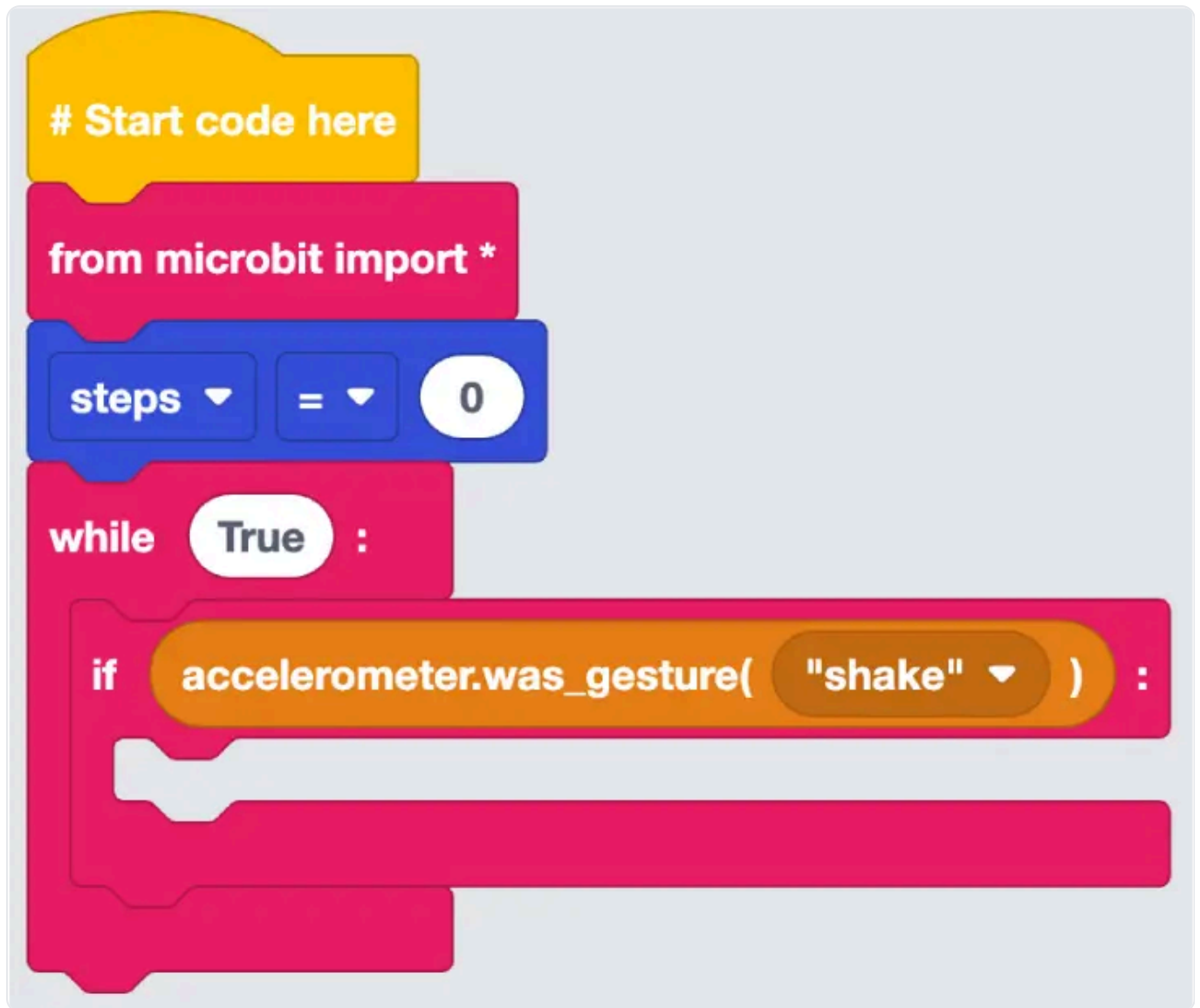
This while loop gives us a space to put code that repeats forever.



8 Detect a shake

1. Select **Basic**, then **Logic**, then drag an `if True:` block inside the `while True:` block.
2. Select **Accelerometer**, then snap an `accelerometer.was_gesture("shake")` block onto the `if` block where it says **True**.

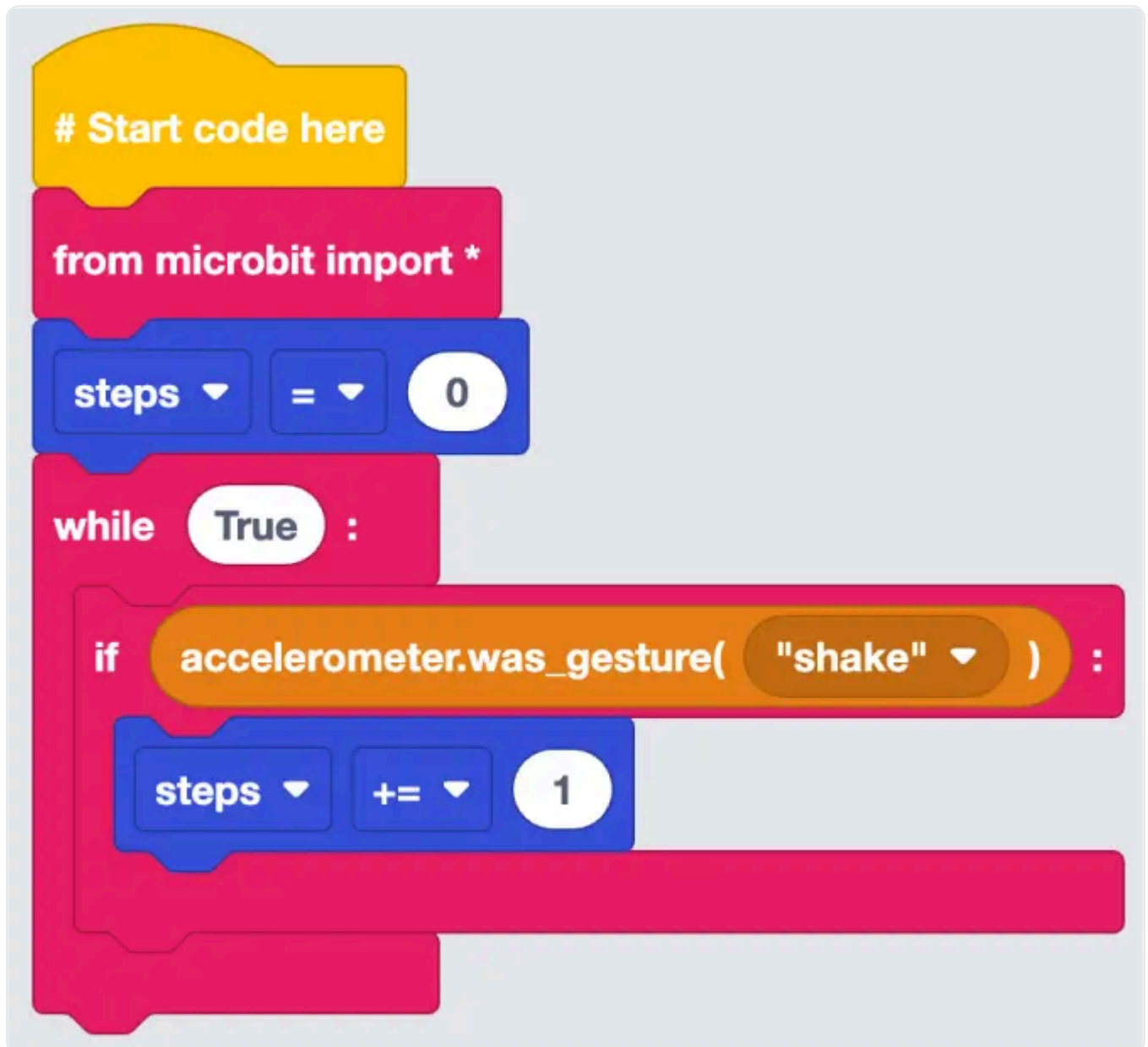
This if statement runs its code only when the micro:bit feels a shake.



9 Increase the steps

1. Select **Variables**, then snap a `steps = 0` block inside the `if accelerometer.was_gesture("shake"):` block.
2. Change the `=` to `+=` and the `0` to `1`.

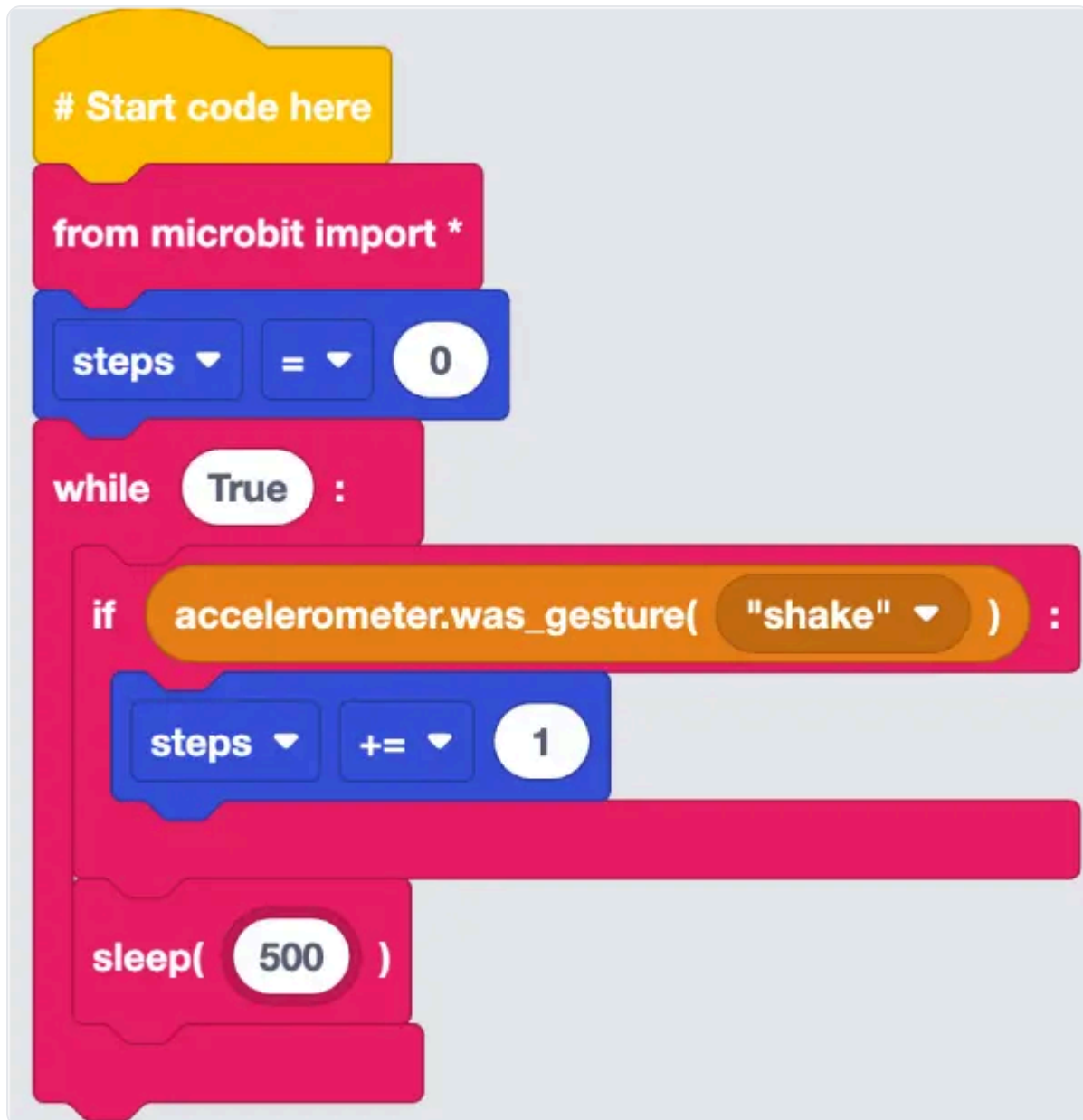
Now every detected shake adds one to your step count.



10 Pause the code

Select **Basic**, then **Statements**, then snap a `sleep (1000)` block **below** the `if accelerometer.was_gesture("shake"):` block, lined up with the `if` (inside the `while True:` loop, but not inside the `if`). Change **1000** to **500**.

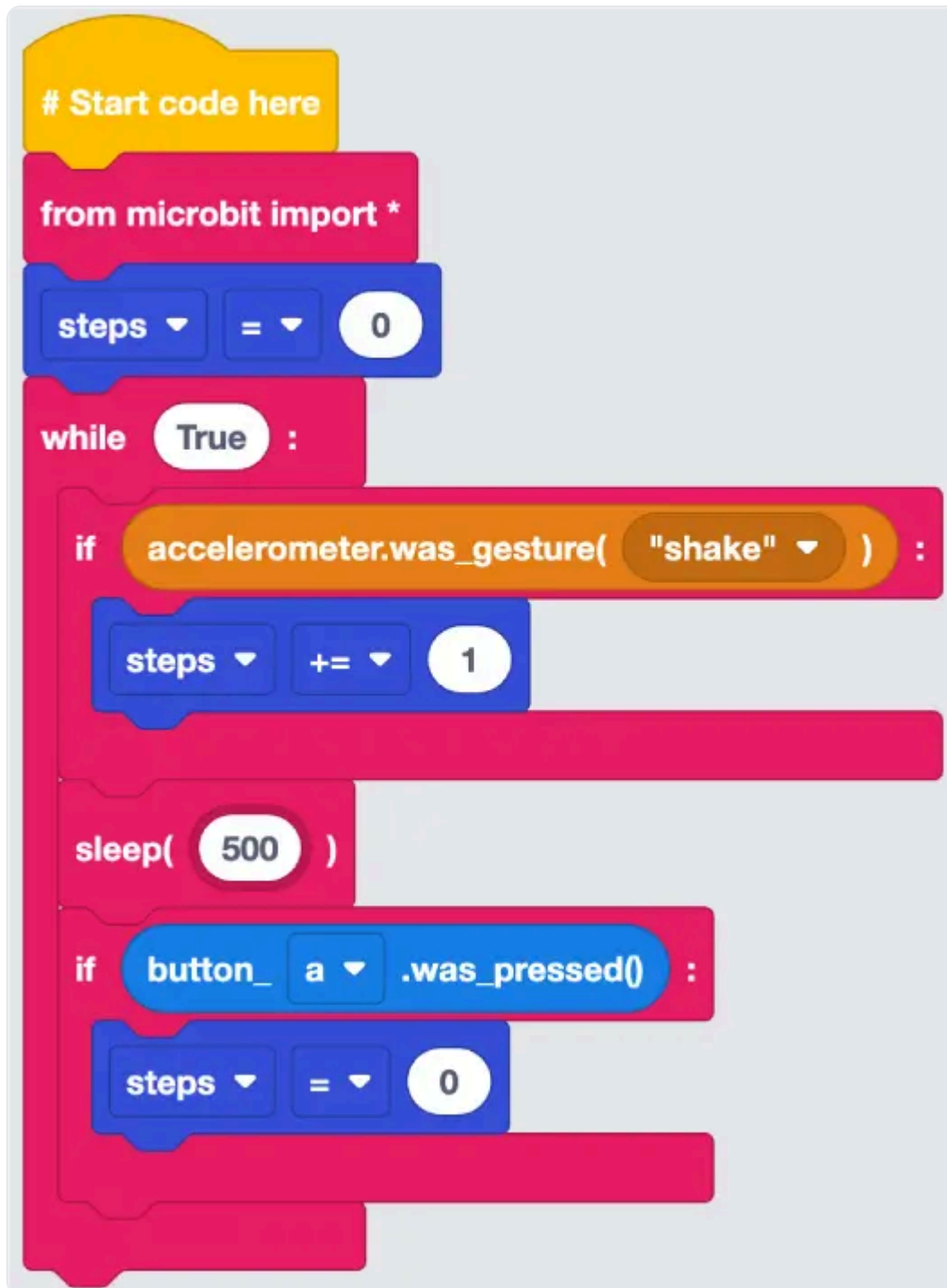
This pauses for half a second so the loop has time to notice the button press we'll add next.



11 Reset the steps with button A

1. Select **Basic**, then **Logic**, then snap an `if True:` block onto the `sleep (500)` block.
2. Select **Buttons**, then snap a `button_a.was_pressed()` block onto the `if` block where it says **True**.
3. Select **Variables**, then snap a `steps = 0` block inside the `if button_a.was_pressed():` block.

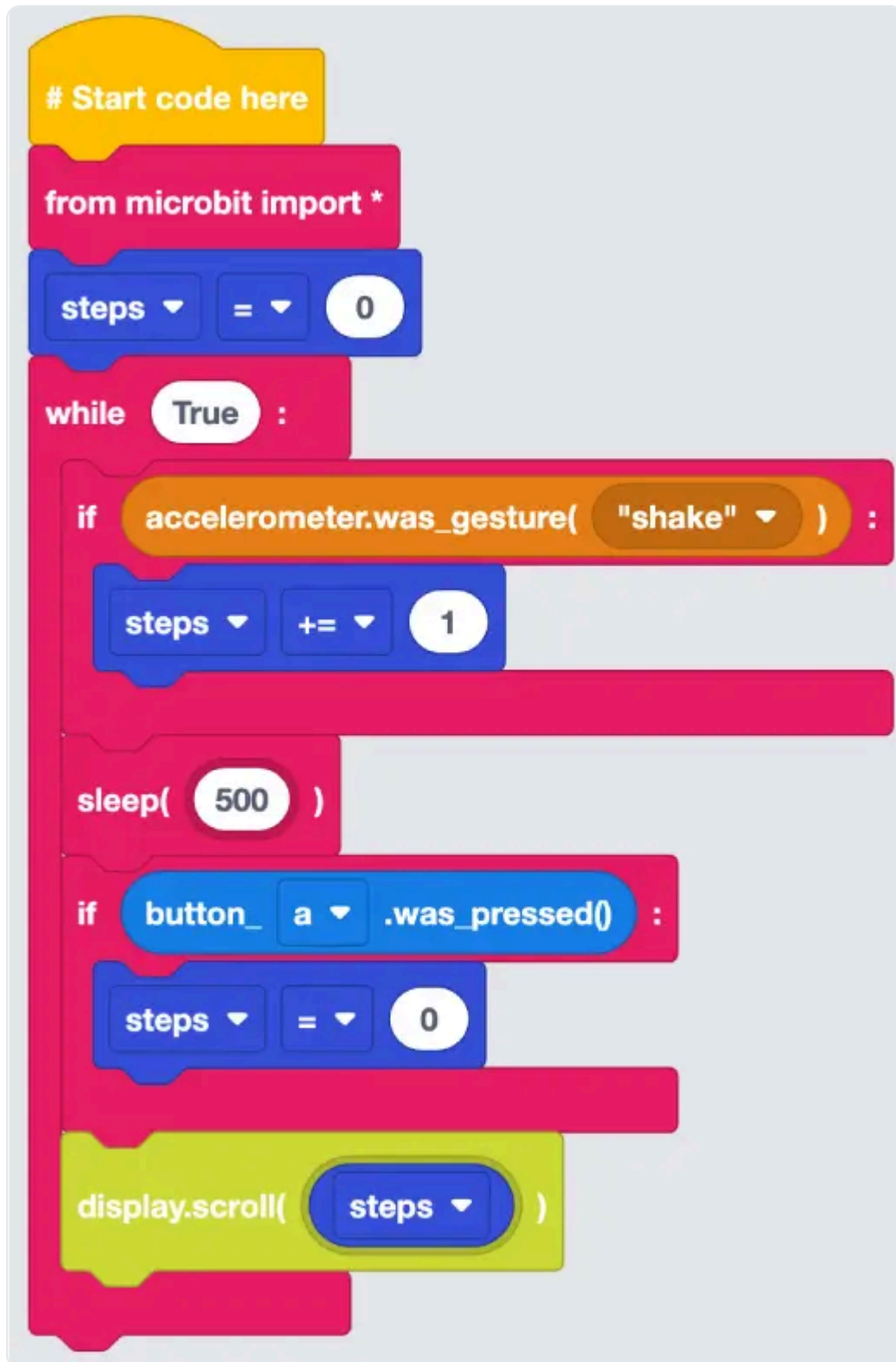
Pressing button A now resets your step count back to zero.



12 Show the number of steps

1. Select **Display**, then snap a `display.scroll ("Hello World")` block just below the `if button_a.was_pressed():` block.
2. Select **Variables**, then snap a `steps` block onto the `display.scroll` block, over the words **"Hello World"**.

This block sits inside the loop but not inside the button A block, so your steps scroll across the LED matrix all the time, not just when you press A.



13 Downloading the Code

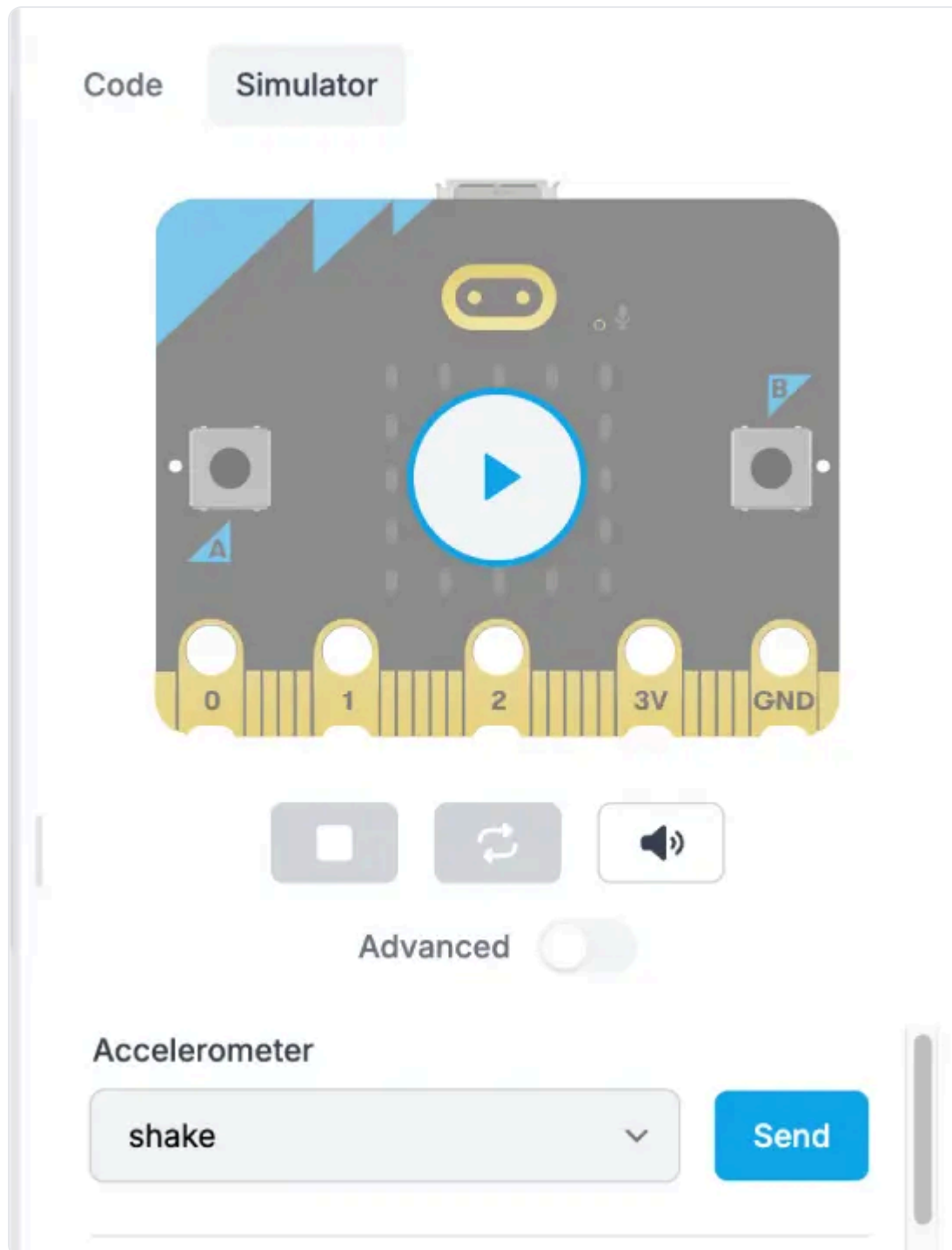
1. Connect the micro:bit to the computer with the micro USB cable.
2. Select **Connect** and follow the on-screen prompts to pair the micro:bit with your web browser.
3. Select **Download** to send the code to your micro:bit.

14 Run it and Watch

In the web browser

EduBlocks has a built-in micro:bit simulator, handy if you haven't got a micro:bit to hand.

1. On the right-hand side of the screen, select **Simulator**.
2. From the accelerometer menu below the simulator, select **Shake**.
3. Select **Send**. This simulates a shake and shows a number on the simulated micro:bit.



On a real micro:bit

Once your code's downloaded, give the micro:bit a shake. Each shake bumps your step count up by one on the LED matrix. Pop a battery pack on, attach it to your wrist or ankle, and head outside to watch your steps climb.

15 Try it yourself

Try it yourself

Challenge: Set yourself a step goal! Add another `if` block inside your `while True:` loop that checks when `steps >= 20`. When you hit it, use a `display.show` block to show a happy face to celebrate, then press button A to reset and go again.

16 Stuck? Quick fixes

Problem	Try this
The number doesn't go up when I shake it	Check your <code>steps += 1</code> block is inside the <code>if accelerometer.was_gesture("shake"):</code> block. Give the micro:bit one firm, clear shake.
The count jumps up by lots at once	A big wobble can look like several shakes. Try one single, definite shake, and check the <code>sleep(500)</code> block is inside your <code>while True:</code> loop so each shake has time to register.
Nothing shows on the LED matrix	Make sure the <code>steps</code> block is inside the <code>display.scroll</code> block (in place of "Hello World"), and that <code>display.scroll</code> is inside your <code>while True:</code> loop.
Button A won't reset the steps	Check the <code>steps = 0</code> block is inside the <code>if button_a.was_pressed():</code> block.
The number doesn't update while I keep shaking	Don't worry, it's still counting! The micro:bit can't show the new number while it's busy scrolling the last one. Stop shaking for a moment and the latest total will scroll across.
My code won't download to the micro:bit	Use a data USB cable (not a charge-only one), make sure the micro:bit is paired, and try a different USB port.